

The Density of Structure

H.K. Andersen

Department of Philosophy, Simon Fraser University

Burnaby, British Columbia, Canada

handerse@sfu.ca

ORCID 0000-0002-7950-8186

Abstract: Realist metaphysical views often rely, explicitly or implicitly, on variations of the presupposition that genuine structure is sparse; call this assumption Sparsity. This includes views where there is one uniquely correct way to carve up the world, and also apparently pluralistic views that allow more structure yet still add a limit so there is not 'too' much. This conflates the question of how to characterize *what* structure is, with two other questions: *how much* structure there is; and *which particular* structures there are. I draw on Dennettian patterns, combining deflationary realism with profligacy about real patterns, to demonstrate a pragmatist metaphysical approach to identifying structure without begging the question of how much structure there is. Pragmatist pattern metaphysics allows that the world may be dense in structure, beyond any presupposed particular upper bound from Sparsity. Finding genuine structure may be easy, and most of it reiterative, useless, or uninteresting; the world may be so dense with structure that we could never exhaust it, such that trying to find 'the' correct structure is in principle unachievable. This shifts the substantive task away from the metaphysical question of what structure is, and towards methodological considerations. If there may be more genuine structure than we could ever investigate, our interests as inquirers must play a role in narrowing down which structures we choose to investigate, and which of many closely related structures yields the most systematicity, usefulness, or other epistemic desiderata. This pragmatist approach is minimalist with respect to metaphysical commitment: saying some structure is genuine says very little of substance about it.

1. Introduction

There is a common presupposition buried in the discussion among philosophers of science and metaphysicians, made by even those who appear to otherwise starkly disagree, about how much genuine structure the world contains. Call this assumption the Sparsity of Structure: that there is structure in the world, but only so much of it, and structure is sufficiently sparse that some or most of what appears like structure is instead merely apparent and not genuine. This is often used as part of an initial set of constraints on what we are looking for in a good account of what structure is, or how to distinguish 'real' from merely apparent structure. This paper will disentangle two metaphysical questions that require different resources to answer, and provide a pragmatist metaphysical answer to each.

This pragmatist approach relies on the inextricable interconnections between metaphysical constraints or principles on one hand, and methodological norms and guidance on the other. It should not be the job of our metaphysical criteria for structure to decide in advance how rich the

world is with that structure. I will offer a brief argument in favour of the possibility that the world may be so dense with structure that it is inexhaustible by description; this highlights the potential danger in presupposing any degree of sparsity in order to yield the conclusion that some apparent structure is spurious or merely apparent. Even if one does not want to take a pragmatist approach to metaphysics, unearthing and rejecting the Sparsity of Structure presupposition is still useful, as it illustrates how a substantive empirical assumption can be baked in as metaphysics.

A modified version of Dennett's pattern realism provides a pragmatist metaphysical foundation for identifying structure. When using pragmatist pattern ontologies without the Sparsity assumption, we end up with more identifiable structure in the world than allowed by other authors who rely on patterns, such as Ladyman, Ross and Collier, and yet with more actual structure, rather than noise, than the version of patterns offered by McAllister. The use of 'pattern' and pattern ontologies as an account of what structure is provides a minimal metaphysical basis from which to proceed to empirical questions like which structures there are. Pattern metaphysics alone yields vast quantities of pattern structure that could be defined, some of which may not be identifiable in any part of the world; it also yields a surfeit of genuine structure because there is a vast array of ways in which any given pattern identification conditions could be slightly re-described, and patterns are defined by their identification conditions. Shifting the identification conditions, including noise tolerance, for a given pattern results in a new pattern; if the first had identifiable instances in the world, the next will very likely also have some. All of those variations can be picking out genuine structure. The world may be dense with such structure.

Using Dennettian patterns means that all of those are equally real, on an extremely minimal sense of realism. In this regard, the pragmatist metaphysical picture shares a surprising amount with Lewis' point about being a realist about possible worlds. Lewis offered realism about one *kind* of thing, namely, worlds. There are a lot of worlds, but only one metaphysical kind, worlds. Here, the pragmatist metaphysician just needs the Dennettian 'pattern'. Then, there are staggering large numbers of possible patterns, and a smaller but still rather staggering amount of those possible patterns that can be used to identify actual instances in the world. There are lots of patterns, but metaphysically, just 'pattern.' This moves the substantive work done away from the purely metaphysical, and into the methodological, including the sciences. This remains metaphysical because we don't have to use every possible variation of any pattern; we use further considerations, such as cohesion with other patterns in an ontology, and coherence among use across different ontologies (e.g. Andersen 2026) Most of the work in picking out structure is not finding *all* the structure that could be found, but rather, finding the best version of those variations, where 'best' is given by further conditions that are not metaphysical.

2. Disentangling two questions about structure

A lot of ink has been spilled in recent decades about how to find structure, and how to distinguish structure that is better in some way - more real, more fundamental, more natural, more scientific, more causally efficacious - from merely apparent but un-genuine structure. Answers have ranged

widely over what structure is, and, accordingly, how we ought to find such structure, especially how to discern the genuine structure from the spurious or merely apparent. There have been, along with these discussions, further discussion about how much structure there is, though these have tended towards the sweeping generalizations of "not very much, that's why it's special" to "a lot, if we find it this way" to "even more than you thought" styles of pluralism. But, as I will argue in this and the next section, many of these views share an underlying assumption about how much structure there is, namely, that there is some upper bound on it such that there isn't too much (for different understandings of too much) and thus it is a central metaphysical task to sort it into real vs derivative or variations thereon.

A quick caveat about terminology. I will aim to present an extremely broad and inclusive understanding of realism, as holding that there is anything genuinely 'out there' such that we are getting at any of it with anything in the sciences. I will be using the term 'structure' for this, in a Sellarsian broadest possible sense of that term. Structure is useful for a couple reasons. It covers a lot of more specific projects, such as those of finding causal structure, investigating social structures, realism about mathematical structures, and more. The word structure does not have to carry any particular significance, other than that there is something 'out there' that our models, theories, discourses, and other representational devices may or may not succeed in picking out.¹ I will also be relying on a pattern metaphysics, with patterns and pattern ontologies, as an account of what structure is; on this approach, other potential specific views of structure, such as entities or invariant mathematical representations, are special cases of patterns, not competitor alternatives. The patterns metaphysics is helpful for illustration but the main concerns about the Sparsity assumption, its consequences, and what it looks like to reject Sparsity, still stand even if one does not want to adopt a pattern-based or other kind of pragmatist metaphysics.

The first question to disentangle is that of what makes any particular structure real, such that an answer to it will tell us what is 'really' real, from that of how much real structure there is. There are many varieties of realism that answer this first question. Answers to this also include accounts of kindhood where some kinds are natural and some are not (though, some accounts of natural kinds also diminish the importance of that boundary; inter alia, Chakravartty 2023, Khalidi 2018). It includes accounts that otherwise differ widely, such as Sider's (2013) reference magnetism where claims about the physical world are made true by snapping like magnets onto the closest uniquely real joints in the world, alongside weak accounts of realism where to be is to be the value of a bound variable (e.g. Quine 1948). It also includes accounts where the question of what is real is left behind in favour of questions about comparative fundamentality (e.g. Schaffer 2009), since this still requires sorting apparent structure into uniquely correct orderings with respect to fundamentality, and still presupposes a high degree of sparsity to what is 'really' fundamental.

¹ Following Price's (2011, 2013) discussion, I avoid 'facts' as awkwardly trying to be both in the world and formulatable in some neutral and uniquely correct way. As Price points out, facts are always formulated in some discourse; there is no discourse-free fact.

Philosophers of science have developed accounts to answer the question of what makes any particular structure real that start from the assumption that our best current scientific theories or models involve real structure, and then 'read off' from those theories what there is. Outsourcing this question to the sciences to determine what structure is genuine is called naturalizing metaphysics (inter alia, French 2014; see Guay and Pradeu 2020 for discussion and critical remarks on this). Various flavours of structural realism construe structure as continuous through theory change; stronger versions such as ontic structural realism also include some forward-looking goal of structure to which the sciences will eventually approach (e.g. French 2006, French and Ladyman 2010, Ladyman Ross Spurrett and Collier 2007).

Some but not all of these answers to the question of what there is presuppose uniqueness of structure. Uniqueness means there is one uniquely right way to carve up the joints of the world, and our job is to figure out what that is, such that our metaphysical analysis is the technique for correct carving and it delivers the correct answer when applied to the world (and merely epistemic limitations might stand in the way). This applies to all the varieties of answers, not just realism: views about fundamentality or grounding also often presuppose that there is one uniquely correct ranking of comparative fundamentality. Some answers, though, don't require the presupposition of uniqueness, no single best way of carving up structure. Cartwright's dappled world (1999) is a realist view that accommodates a wide plurality of structure; similarly, rainforest realism, and other similarly named realisms (e.g. Dupre 1981). In these cases, there can be more than one way of carving things up, even for the same set of phenomena, and a plurality of ontologies can co-exist.

Answering this first question, about what there is, involves some non-zero level of realist commitment but can vary with respect to how unique assumed results of applying that to the world will be. In addition to varying with respect to uniqueness of structure, these views can vary along a second parameter in terms of how much strength is involved in the realism itself. The strength of realism, this second parameter, can be extremely robust, such as the knee-jerk realism of Sider (2013). It could also be extremely minimal, such as Dennett's pattern realism (1991). Weak realism can allow for a larger range of possibility for how much genuine structure there could turn out to be, because there is much less involved in committing to the reality of such structure. If saying, yes this is genuine structure, carries comparatively little import one can allow in more structure. If it is a very weighty and serious commitment, one might be more selective in what gets to count as real. In this regard, uniqueness of structure and strength of realism are part of answering the question of how to identify genuine structure, or, to differentiate genuine from merely apparent structure.

The first question, what makes any particular structure real, can thus vary along at least two parameters, the uniqueness of ways of carving up the world with that analysis, and the strength of realism attributed to the results of applying it. The second question to disentangle, then, is that of *how much* structure there is. This second question, how much structure there is, is easily conflated with the first question, what *is* structure, because the second question is closely connected to, though not the same as, this second parameter along which the first question varies. If one increases the weightiness assigned to anything that is real, on a given account of realism, then this might lead one

to (though it is not necessitated to) presuppose that, because it is so weighty to call structure real, there oughtn't be much of it.

It may be that we live in a world that has very little genuine structure (though, in my view, there is already sufficient evidence to tell strongly against that). It may be that we live in a world that has lots of structure, like a dappled or rainforest realism. But, as I show in the next section, even these views tend to add some upper bound to how much there could be, such that there are amounts that seem too high and are ruled out in advance.

Answers to the first question can thus implicitly presuppose some acceptable range of answer to this second question, and in doing so, presuppose an overly restrictive range. This presupposed range of acceptable quantity of structure, or, answers to the second question, informs answers to the first question, which aim for a characterization of what structure is that will necessarily yield less structure when applied as criteria to empirical investigations. In presupposing some sparsity, even if it is comparatively generous, these views add a substantive and conceptually prior metaphysical presupposition, as an implicit answer to this second question. Both the spare and the profligate ontologies may involve a priori metaphysical commitment that thus shapes subsequent inquiry. This expectation of comparative sparsity rules out a priori the possibility of being surprised by how much genuine structure a given view of realism might turn up when applied in the sciences.

In the next sections, I'll urge that we should not presuppose that there is 'only so much' structure prior to finding out what structure there is and what the best methods are for distinguishing structure. In addition to being an unjustifiably strong initial assumption, Sparsity can also lead to a misconstrual of the task of developing ontology from the sciences using metaphysical criteria, in two mis-steps. One mis-step is to presuppose that there must be one uniquely best or correct ontology. The other mis-step is to construe the task of finding 'the' structure as one where we should be trying to find *all* the structure: by finding the way to identify genuine structure, we can find all there is to be found, and thus complete the task of differentiating structure from non-structure. It might turn out that structure is sparse, but we ought not presuppose that in advance. A streamlined metaphysics can answer the first disentangled question and then treat the second question as *empirical*, not *metaphysical*.

3. How much structure is there? A range of answers

In order to further disentangle the questions of what structure *is* from *how much* of it we should expect to find, this section compares some specific answers to the question of how much structure there is, in order to show how the assumption of Sparsity can show up even in profligately realist views. I'll start with the strongest versions of Sparsity, and then contrast three different views that rely on patterns metaphysics of structure, showing how Sparsity shows up even with apparently inclusive accounts of realism.

One end of the spectrum with respect to sparsity of structure is the view that there is one and only one correct set of joints in the world (Uniqueness), there are not many of those special joints (Sparsity), and the goal of science is to eventually identify them all. This is common in contemporary post-Lewisian analytic metaphysics (e.g. Sider 2013, North 2021, Schaffer 2009). There may be more than one set of joint-like structures, such as in the grounding literature, but even if the chain of grounding goes on indefinitely, there is still a unique ordering of fundamentality. It is easy to see how these views presuppose sparsity of structure. They also offer a realism that weights the second parameter especially heavily: it is a very big deal, metaphysically, to label a joint in the world as genuine, so we shouldn't do it too often. Here, we can also see, from the presupposition of both Uniqueness and Sparsity, that it makes sense to then think of the task of the sciences as that of finding all the joints in the world (Totality) such that there is some Final Science that will deliver those. It is fairly common in post-Lewisian metaphysics to think it is at least possible for there to be an End Science such that the contents of that end science can be referred to, even if we don't yet know what they are. The intuition that metaphysics is not just about finding genuine structure, but also *all* the genuine structure, then drives the intuitions of Uniqueness and Sparsity, without which Totality would be unachievable.

One might want to oppose strong views like knee-jerk realism to views like rainforest realism, offered by Ross, Ladyman, and Collier (2007) (see also Ross 2000 for a different version), which rejects both Totality and Uniqueness. They draw on Dennett's patterns with modifications to the definition of patterns. Their view of realism rejects objects as the basis for ontology, and allows in so much as real that they liken it to the rainforest in contrast to a Quinean desert landscape. Even here, though, there is an implicit reliance on some upper bound of structure such that Dennett's original view allows in too much and consequently must be modified to avoid that.

...let us return to Dennett's account of real patterns. There are, as noted above, problems with it. Though it offers the major achievement of the compressibility requirement—that a real pattern cannot be a bitmap of its elements—as a *necessary* condition on real-pattern-hood, Dennett never provides an analysis of *sufficient* conditions. (Obviously compressibility isn't sufficient, or every redescription of anything would conjure a new pattern into being.) (p. 205)

This parenthetical remark at the end conveys the implicit metaphysical assumption that there is only so much structure. They justify the modification of the definition of structure (in this case, patterns) as required to limit the amount of structure it allows for. On Dennett's original version, a redescription of a pattern changes either the identification conditions for it, or the noise/error tolerance, such that redescription would, in fact, pick out a mostly overlapping but non-identical pattern. This is ruled out by Ladyman, Ross, and Collier as allowing in too much structure; they attribute almost magical properties to redescription as conjuration of structure into existence (this issue of redescription of patterns will figure centrally in the next section).

The metaphysical criterion by which structure is identified, the definition of patterns, is restricted in advance of using that criterion to see how structure there is. This move presupposes a much higher upper bound on the second question, how much structure there is, than views with sparse and

unique joints, but still presupposes a bound that is lower than what the original patterns view offered. And, importantly for my purposes, it presupposes the answer to the second question as part of giving an answer to the first question.

Consider a view on the furthest other end of this spectrum, that of McAllister (especially 2010, 2011; see also 1997). McAllister draws on patterns and the distinction between data and phenomena (Bogen and Woodward 1988). He argues that all possible patterns can be found in any data set, allowing for any given pattern to be instantiated with 100% noise. Since structure in datasets reflect phenomena, or, structure in the world, this means that the world exhibits all possible structures, using a noise/error tolerance up to and including 100%.

Any empirical data set exhibits all possible patterns, each with a certain noise term. The magnitude and other properties of this noise term are irrelevant to the evidential status of a pattern: all patterns exhibited in empirical data constitute evidence of structures in the world. Furthermore, distinct patterns constitute evidence of distinct structures in the world. It follows that the world must be regarded as containing all possible structures. (McAllister 2009, p. 73)

This handily defines the other end of the spectrum, as far from the commitment to uniqueness of structure as possible. It answers to the first question, what structure is, and adds an empirical component to the second question, how we ascertain how much structure there is, by the use of datasets and the patterns in data as evidence for structure in the world. Rainforest realism reins in patterns as too profligate, while McAllister's version goes the entirety of the other direction, allowing in all possibly definable structure as genuine. McAllister has reasons for taking this consequence on, including an emphasis that defining structure doesn't conjure anything into being. For my purposes here, though, if the world is stipulated to contain all possible structures before we even investigate it, then the notion of structure in the metaphysical analysis differentiates nothing when applied to the world, thus not accomplishing the task it was meant to for realism.

With knee-jerk realism and a sparse unique set of joints on one end, and all possible structures on the other, the full range of possible outcomes to the question, how much structure is there? can be arranged. Rainforest realism uses patterns, which rejects Uniqueness, but reins in how many patterns there could in principle turn out to be, thus capping the answer to this second question before empirical investigation. This is a form of the Sparsity assumption.

A third way of using patterns focuses on causal structure specifically, but it is useful to illustrate a realist use of patterns more inclusive than rainforest realism yet more discriminating than

McAllister's view.² In Andersen (2017), I offer an account of causal structure³ that draws on Dennettian patterns, where causal relata are patterns identified in the causal nexus (a technical term). Causal variables in interventionism, on this view, just are ways of picking out patterns in the causal nexus. They involve giving the identification conditions for instances of the variable type, including (though often implicitly) a noise tolerance that covers both the possible error associated with measurement and the noise that could be associated with physical instantiation separately from measurement. A variable thus picks out a set of tokens in the world that count as instances of that variable taking one of its values. In this approach, to redescribe a variable with slightly different noise tolerance different identification conditions *is* to pick out a slightly different pattern; the different variable versions will have overlapping but non-identical set of instances. At most one such variable-variant can be used in any given system of variables, but there is no single uniquely correct, or complete, system of causal variables. Empirical considerations related to coherence or cohesion (Andersen 2026) are used to narrow down which variant of identification conditions plus noise/error should be used for a system of variables under particular modelling conditions, not metaphysical considerations about getting 'the' causal structure.

Thus, on this use of patterns, changing the description of the pattern slightly does indeed change the pattern being picked out. Contra Ladyman, Ross, and Collier, though, this does not "conjure a new pattern into being" in the sense of forcing the world to contain something that it didn't prior to redescription. But, of course, the original pattern did not conjure anything, either.⁴ All that pattern identification conditions offer is a well-defined pattern of which we could check for instances; it doesn't ensure there are any instances. That there are descriptive variants of the same pattern doesn't mean that one of them is already wrong and should be ruled out a priori by metaphysical definition. Both (re)descriptions might have identifiable instances in the world, or one and not the other, or neither. Further considerations than patterns alone are required to decide which variant is more apt to use. Pattern metaphysics does not require any particular limit on how many patterns we are

² There are other accounts that involve pattern metaphysics that I am not including here. For example, Petersen (2019) and Millhouse (2022) both rely on patterns for realism. The view of patterns I develop here differs from theirs not along this spectrum of how much structure there is. Instead, they share a different assumption I reject for other reasons, namely, that the world has some kind of pixelated or pixel-analog structure directly akin to the Game of Life. There isn't space to address this here, but it does conflict with Density as developed in the next section.

³ It is important to note that not all identifiable patterns count as causal relata; it is only patterns identified in a certain substratum (plus other conditions). There could be other patterns instantiated such that they count as structure but not causal structure.

⁴ The authors seem to be concerned about how our definition of a pattern could shape which patterns we subsequently find. There is a very useful distinction in Chang (2022, especially chapter 2) between mind-framed and mind-controlled that alleviates this concern. Patterns are mind-framed, in that we choose the concepts by which we frame them; but there is no getting away from this, as any other pattern we choose would also be mind-framed. This is a variation of the point made by Price (2011, 2013) that no matter what fact we formulate, there must be *some* discourse in which it is formulated. What Ladyman et al seem to want to avoid is having patterns that would be mind-controlled on Chang's clarification of mind-dependence. In this regard, they need not add any disclaimers or make changes to the original definition: all patterns, even those slight redescrptions, are not mind-controlled. We cannot make the world do what we want just by changing our definitions (as convenient as that would be). Even for Hacking's looping kinds, it is not the pure redefinition of a kind that changes the world; it still goes through causal connections in the world to shape, and cannot be controlled or shaped in the exact way we want.

allowed to define, and can't magically make new patterns appear in the world simply because they have well-defined identification conditions.

McAllister, on the other hand, offers a more inclusive answer to the second question, how much structure there is, without adding any prior limit to patterns, by including all possible structure as findable in the world, not just all possible patterns as well-definable enough to look for. This answer to the second question--how much is there? all of it--still forces a specific response to the first question. The plenitude of structure is built as a metaphysical posit into the definition, rather than defining structure and leaving it an empirical question about how much structure the world contains.

The use of patterns in the foundations of causation account (Andersen 2017) falls between these two on the spectrum of how much structure there could be, holding fixed a very mild realism. It relies on the Dennettian view of patterns for the metaphysical definition of structure, but leaves open which of the possible structures we can define and look for are actually found in the world. Most importantly, it leaves open how much structure can be found; slight variations on the same pattern could also turn out to successfully identify structure. Some patterns will be well-defined but also turn out to have an empty set of instances, contra McAllister. But a vast range of patterns that are closely related to one another could all turn out to successfully identify *some* structure in the world. This is what a patterns metaphysics for what structure *is*, that does not presuppose an answer about *how much* structure there is, looks like.

4. The Density of Structure

An immediate response to this profligacy with respect to patterns is that this is far too much structure, much of it redundant in the sense that slight redescription pick out overlapping pattern instances, not genuinely independent and 'new' structure. One might object that even if one agrees that we should disentangle the conflated questions of what structure is, and how much of it there might be, there must be some limit to how much structure there is, and that limit must be lower than what turns out as possible on patterns metaphysics. This objection might press further: surely these two questions can be disentangled without thereby being completely separate. Answering either requires some commitment with respect to the other.

I agree that these questions, while distinct, involve one another. Once one commits to a view on what structure is, that thereby constrains how much of it could be found, and vice versa, where if one holds that there should only be 'so much and no more', then this limits what metaphysical views could be consistent with that. Nevertheless, there are at least two good reasons to leave, in the metaphysical characterization of what structure is, as wide a margin as possible for empirical answers to how much structure there is. One reason is that presupposing any upper limit to how much structure there is unnecessarily inflates the metaphysical by deflating the empirical or epistemological. This will be developed in the next section. Another reason to leave this wide margin is to avoid accidentally getting it wrong: if the world does turn out to have more structure than expected, we would be systematically prevented by our own metaphysical commitments from

discovering this. This will be developed in this section, through an argument that a patterns metaphysics is both minimal and also leaves open that the world is so dense with structure that we could not exhaust it via description. Even if we defined new patterns forever, and checking the world for them took no time, there are patterns with continuously varying parameters. Variations of them exist for every point on a real number line, which is uncountably infinite, while our ability to define and check for patterns is somewhere below countably infinite. We simply couldn't check for all possibly definable patterns. There wouldn't be any upper limit that could conceivably count as sparse.

The procedure for using pattern metaphysics for structure illuminates this. Consider pattern metaphysics with no added stipulation about how much structure there may or must be. Start with some pattern A, with determinate identification conditions, along with a specified noise tolerance less than 100%. This means that pattern A is well defined, by which I mean, one can pick up some chunk of the world and ask of it, does pattern A occur here? and be able to determinately return a yes or no answer.

For a pragmatist metaphysician, the metaphysical part of the task is pretty much over. The work of pragmatist metaphysics was to provide a characterization of what structure is, such that one could empirically apply that notion to identify structure in the world. The metaphysical tasks did not include specifying which precise patterns one should use. The pragmatist metaphysician could find pattern A, then decide to refine the identification conditions for pattern A slightly, into pattern A', and find instances of pattern A' as well. The metaphysics of 'pattern' just provides the bare schemata of structure. It does not specify which structures one will find, nor which to go looking for; those remain empirical questions. Importantly, the metaphysical definition of structure in terms of patterns does not provide the resources to decide which of A and A' are 'the' correct pattern for the structure, because both have successfully identified structure. Both are, minimally, real.⁵

It is entirely possible for both pattern variations to pick out genuine structure in the world. In the causal patterns example of the previous section, slight variations on causal variable identification conditions might result in slightly different instances of the variable taking one of its values. But all of those might be genuine causal structure. The bare definition of what a variable is allows for this variability, because that is how systems of variables can be fine-tuned together, as a system. It would violate conditions like Causal Markov if we were to include both of those variants in the same system of variables. But each could stand in a causal relationship to some other, non-overlapping, variable, in two different systems of variables. They each pick out genuine, mostly overlapping, causal structure in the world; the further question of which version we should use is a modelling

⁵ This paper is focused on pattern metaphysics for a pragmatist philosophy of science, and thus focused on the role patterns can play in scientific practices and theoretical units such as models. Patterns are, of course, also very useful for non-scientifically oriented patterns. Dennett originally developed patterns for agency and intentions; it is an advantage of patterns not explored in this paper that pattern metaphysics does not privilege scientific patterns over any others as intrinsically realer.

question.⁶ Once a system of variables has been defined, it becomes an empirical question if there are instances of those variables, and if there are causal relationships among those variables that reflect causal structure connecting the instances. Empirical considerations and modelling goals shape which of many variations on a pattern should be used. Having defined them doesn't ensure that they thereby are real, even on this minimal, deflationary view. That any of the possibilities do pick out genuine structure doesn't mean that we thereby *must* use it, if a variant version is better suited for modelling purposes.

More generally, for causal and non-causal structure, there are so many variations on any individual pattern's identification conditions, especially for any pattern of which an instance has been positively identified, that the world could be dense with variations. Dense is used here the same way as the real number line is dense. There are uncountably infinite variations on any pattern that involves identification conditions with a variable defined over the real numbers. In Andersen (2026) I argue that patterns should be used in ontologies that connect different patterns together cohesively, and which coherently connect different uses of the same or very similar patterns in different usages. Those constraints are not strictly metaphysical, and they are what rule out most patterns. That, not pure metaphysics, is how the density is winnowed.

Even if we did nothing but describe genuine structure from now until forever, we would still not exhaust all the things one could truthfully say about what structure there is. The task of metaphysics cannot be to *exhaustively* describe real structure in the world, if there is a possibility that it is inexhaustible. The main reason that giving an exhaustive description of all the structure seemed feasible is that it was backed up by the Sparsity of Structure assumption. Any metaphysical view that involves the idea of End Science or a Final theory is presupposing Sparsity and from that, inferring to some future time when we have found "it all."

There is some upper bound on structure lower than McAllister's, because there are well-defined patterns of which no instances ever occur. Yet there are more ways to create well-defined patterns than we could ever exhaust. Especially once any minimum time and resources for checking if a pattern has instances are taken into account, there are far more well-defined patterns than we could ever check to find possible instances of. That upper bound might be staggeringly large. It will always relevantly outstrip every pattern we can come up with, because many could be iterated by varying a real-valued parameter. Patterns metaphysics can allow that there may so much structure it would be hard not to find at least some. There may be so much structure that we have so many options in terms of how to model it that we cannot just use the idea of the 'right' structure as a guide. It must be supplemented with other modeling considerations.

Thus, the pragmatist metaphysician should not begin the metaphysical task by presupposing any particular answer, or even a range of answers, with respect to the question of how much structure

⁶ There have been criticisms of the system relativity of interventionism on these grounds (see Parkkinen 2022 and Dong 2023 for different responses to this concern). The resistance to this system-relativity rests on exactly the same underlying presupposition highlighted earlier: that this is somehow too much causal structure, and additional criteria need to be added in order to eliminate or at least alleviate system-relativity of causal relationships.

there is. This is not a priori answerable by pure metaphysical considerations. The answer to how much genuine structure there is might range from extreme sparsity, from which all the rest of derived or on which it is dependent, to density akin to points on the real number line. There is no metaphysical magic that will ensure in advance that there is only a very little structure.

5. A realist pragmatist patterns metaphysics

In this section, I will argue that there is a better way to be minimal about metaphysics by allowing in all sorts of stuff as legitimate structure, and then doing the hard work of narrowing it further using methodological and empirical inquiry. Pragmatist patterns metaphysics follows in the footsteps of Dennett's pattern ir/realism, where there is very little to which we commit when we say some structure is real. In the face of possible density of structure, a pragmatist metaphysician leans heavily on further considerations to direct inquiry: which patterns are worth the time to define, or look for, and meet other desiderata for modelling or prediction or discovery? Pragmatism makes a place for our own interests in deciding what to investigate, not because those interests 'make' the patterns real, but because there is so much that is real and comparatively so little time that we should take all the additional constraints we can find to further narrow the space of possible avenues of inquiry.

The question that should be asked here is not which of some cluster around pattern A of A, A', A*, and so on, is 'the' metaphysically right pattern. The pragmatist can accept that overlapping identifiable structure, because counting as structure doesn't signify much on a weak notion of realism. In general, there will be lots of findable structure, and most of it is pretty useless. What we want, in the sciences especially, is something like genuine structure that also generalizes effectively, or has a lower noise tolerance, or has a higher noise tolerance but is much easier to check for. Having several viable candidates for pattern variations is a great improvement over insisting that there can be only one. This is where the sciences then iteratively refine the pattern identification conditions among potential variants, better fitting patterns together into cohesive ontologies, and fitting those ontologies onto the world.

An empirical example is helpful. Consider Chua's (2021) discussion of slight variations in identification conditions for entropy in thermodynamics and statistical mechanics. Each of these definitions essentially offer a slight variation of a pattern, akin to pattern A, A', A*. Each gets at something important, and is well-confirmed: each identifies real structure. Mostly, physicists can use them without having to be concerned about the ways in which they differ; some definitions are better suited to one physical regime than another. But that does not settle one as the 'right' definition of entropy. Part of what entropy does is connect a series of other patterns in a rich tapestry that involves conceptual connections (such as equations in which the pattern is related to others) as well as experimental practices for measuring in various conditions or regimes. It sits at a nexus of other patterns as well as actions, interests, and other considerations of scientists. It is the way in which these definitions could fulfill these various functions at that particular nexus that answers questions about which version best suits a particular context of use. There isn't a further metaphysical case that one is uniquely correct; they are all correct, for something, and the process of refining the

definitions, bringing them into alignment, is the process by which sciences precisify patterns (see also Robertson and Prunkl 2023).

A similar point can be found in the context of quantum wavefunctions and real ontology as a response by Mulder (2024) to Wallace (2012). Mulder points to the way in which Wallace fixes the metaphysical criteria, via what Wallace calls Dennett's Criterion, in order to reduce the amount of structure that thereby results from adopting Dennettian patterns. Mulder's raises the concern that this unnecessarily restricts patterns that otherwise meet the criterion. This is a further example of how the presupposition of Sparsity, in a patterns context with a comparatively generous upper limit on how much structure there could be, can yet be worked into the metaphysical definition of what structure is, so as to not exceed some a priori upper limit.

Chang (e.g. 2004, 2022) also provides a realist pragmatist perspective that similarly defuses the concern that being able to define more patterns, or, more concepts, somehow means we thereby will them into existence. He repeatedly demonstrates how our cognitive limitations, goals, interests, and historical situatedness play ineliminable roles in the content and structure of scientific products like models, theories, etc., without thereby undermining the epistemic legitimacy of those products. Our languages, including formal languages and mathematics, are mind-framed. This does not, though, mean that these are thereby mind-determined. And, there are no non-mind-framed options to use. We can make up whatever patterns we like, but that does not mean we will actually *find* instances of them in the world, nor does it mean that we can simply will them into being useful, into making good contributions to science, or being applicable outside some incredibly narrow and gerrymandered range of application. The sciences aim to fit mind-framed representations onto the world, which greatly constrains what makes it out of the metaphysical starting gates.

Pragmatism emphasizes methodological considerations such as holism to provide constraints on which structures to investigate. These inquiry- and inquirer-specific considerations must be involved when there is so much potentially findable structure: we only have so much in terms of cognitive resources, time, and so on. We *have to* decide what we want to focus on. There are frequent misunderstandings of pragmatism as thereby putting human-concerns ahead of more 'objective' considerations. There is a suspicion that being a pragmatist and emphasizing the role we play in choosing what patterns to use, refine them, and investigate, compromises the resulting science. These human-framed considerations are sometimes dismissed as merely pragmatic. This is a misunderstanding. Being a pragmatist is not the same as being pragmatic; the pragmatist has effective metaphysical criteria for finding structure. Pragmatist metaphysics doesn't offer methodological and conceptual considerations *in lieu of* finding genuine structure. It offers these considerations *in addition to* finding genuine structure, because they are needed to further focus our epistemic efforts. The task of finding genuine structure still leaves us with far too much that we could find, a potentially inexhaustible richness, most of which is not especially helpful in science. We have to invoke other considerations in order to proceed, and are better off being upfront and explicit about this, because it improves how we do it.

It is simply false to think that 'The Right (Sparse) Structure' is 'truly' objective, so that involving considerations about our own interests and goals involves compromising objectivity with human subjectivity. We already have to make these choices. It is a false dichotomy to claim that *either* we get at real structure, *or* we compromise, in a merely pragmatic (not pragmatist) way. We can acknowledge that our limitations and interests play an ineliminable role in our investigations of structure. A fully pragmatist metaphysics is both/and: the sciences can get at real structure with patterns, for example, *and* our interests and limitations still must play an ineliminable role because not all real structure is equally useful and it is impossible to exhaust all the structure that could be found.

Disentangling these questions about structure acknowledges this interconnection and streamlines the part involving metaphysics, so that it does not attempt to provide a priori answers to questions better left to the sciences. Patterns provide one way of answering this question, surprisingly similar to a point made by David Lewis and realism about possible worlds. Lewis' reasoning was that one only needed to be a realist about one *kind* of thing, namely, worlds. There were a lot of them, but still one kind of thing. The ontology list of "what there is", involving all the worlds, was long, but the criterion used to decide what got on that list was very minimal. Pattern metaphysics is similarly streamlined. There are many different kinds of patterns that can be defined, and many instances in the world of well-defined patterns, such that we might have a lot of different ontology lists and they might get quite full. But the *metaphysics* is minimal: the criteria to get on an ontology list is 'pattern', with identification and noise tolerance specified; the rest is empirical.

The pragmatist metaphysician then differs from a post-Lewisian analytic metaphysician. The latter might attempt to develop additional criteria, such as fundamentality or naturalness, in order to get further on the question of genuine versus derivative structure. This metaphysical move fuses two questions, using modifications to metaphysical criteria for identifying genuine structure to get into an already-determined correct range of sparsity regarding how much structure there is. The more empirically inclined pragmatist would instead provide a metaphysical answer to the issue of how to discern genuine structure, and then use that to inform explicitly methodological considerations, where the metaphysical would constrain, but only partially, investigation into what structure there. But, importantly, it would not constrain or guide with respect to *how much* we expect to ultimately find, such that finding more is somehow a problematic outcome that should be 'corrected' by changing the metaphysical criteria to result in less structure.

From the other direction, the naturalized metaphysics approach (e.g. rainforest realism) holds that we should read off our metaphysics from our best current scientific theories. Like the post-Lewisian commitment to sparsity and uniqueness, though, this is not minimal in terms of metaphysics. This naturalized metaphysics procedure might get us the fewest items on a list of the furniture of the world. Yet it requires more substantial commitments regarding the metaphysical principle(s) or criteria by which something makes it onto that list, since those differ between different models or

theories. As such, it is minimal *ontology*, but not minimal *metaphysics*. A minimal metaphysics might result in a much larger ontology or ontologies by way of fewer metaphysical commitments.⁷

Trying to constrain the definition of patterns so the sciences turn up less structure offloads more work onto the metaphysics. It might streamline our ontologies, and simplify methodology, but it does so by taking issues that are in fact the proper domain of methodology and putting them into the upstream metaphysics. It also impedes methodology: there is little left to do but rote apply the purely metaphysical criteria, with a recipe that smells suspiciously like the search for "the" method of science. In contrast, streamlining the metaphysics moves that load-bearing role of answering how much structure there is, and which exact structures there are, back into epistemology.

6. Conclusion: Pluralism, Realism, and Pragmatist Metaphysics

Contemporary versions of pragmatism for philosophy of science (inter alia, Andersen and Mitchell 2023, Chang 2022) emphasize the inextricable interconnections between metaphysical and epistemological or methodological considerations. Just as we can't learn from epistemologically isolated metaphysics about how much structure, or which structures, there are in the world, we cannot do pure methodology that involves no metaphysical commitments. If we try, we still end up doing metaphysics, just implicitly and poorly. We can utilize that interconnectivity to improve how we do both by how they mutually constrain one another.

There is a *prima facie* tension within pragmatism between pluralism and realism. Pluralism with respect to different models, theories, ontologies, etc., means allowing for the ongoing coexistence of representations that conflict to some degree. Especially when incompatible models involve overlapping domains of phenomena, there will be competing claims about what parts of the world are like that cannot be neatly reconciled. One realist move is to commit to all but one being ultimately eliminable, reducible, or otherwise inferior, thereby resolving the tension. The embrace of pluralism, to such a realist, seems too close to endorsing contradictions or to giving up on resolution. When it allows for the ongoing existence of partially conflicting models, pluralism seems to reject the very possibility of there being one world behind those representations.

However, the pluralist can also be realist to a modest degree *now*, rather than having to wait to figure out which is the 'right' theory or model. Patterns pragmatism shows how apparently conflicting ontologies can all have found some real structure, without having to decide which one is more 'right', and while still maintaining the underlying unity of the world, because that world is rich enough with genuine structure that these descriptions are all getting at parts of it but will never, even collectively, exhaust it).

This misconception that pragmatism is necessarily, and problematically, pluralistic is closely connected to the view that there is One Right Set of Joints in the world, and that we could in

⁷ See also Mitchell (2023) for another way in which pragmatism reconciles pluralism and realism with a thinner notion of realism, drawing on affordances rather than patterns.

principle find them all and then be done. Realism need not endorse the uniqueness of a single, sparse, way of individuating structure. It is a false dichotomy to assume that by rejecting a single uniquely correct *and sparse* way of identifying structure in the world, one is thereby required to accept that there is no structure, or that any structure is as good as any other, or that is it 'merely' human interests, somehow devoid of contact with that world, that settles questions of structure. A pragmatist can reject the claim that there is One Right Set of Sparse Joints, while also retaining realism. The key is getting the correct direction of fit for involvement of the human-oriented aspects of pragmatism. It does not go *from* us and what we posit or imagine or wish, *to* making the world fit our interests. Rather, we have interests and goals in finding structure, and those must fit onto the world in order to serve those interests. The direction of fit is *from* the world *to* how we can represent and act within it effectively.

This paper thus illustrates one way (certainly not the only way; pragmatists also disagree about this) to reject sparsity and uniqueness while being realist about structure. Research interests must be involved because there is simply more structure that is well-definable than we could ever manage to look for, but this is not to say we can make there be structure because we want it to be there, or because it would be convenient for our goals. Most of that structure may be low quality, and considerations that are not metaphysical provide the further criteria for how we should go about spending limited (as in, not literally infinite) research efforts in the sciences.

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